

## Practice problems

1. Maternal imprinting of *IGF2* gene (i.e. silencing of maternal allele) is a result of \_\_\_\_\_.
  - A. methylation of histones in nucleosomes
  - B. methylation of enhancer sequences
  - C. nonmethylated insulator
  - D. methylated insulator
  - E. nonmethylated enhancer
2. A mutation in human cultured cells results in no expression of drosha. Which of the following RNAs will be accumulated in this cell line?
  - A. All mRNAs
  - B. miRNAs
  - C. pri-miRNAs
  - D. pimRNAs
  - E. siRNAs
  - F. pre-miRNAs
3. In general, which of the following modifications make chromatin less compact and more accessible to transcription? Check all that apply.
  - A. Histone acetylation
  - B. Histone deacetylation
  - C. DNA methylation
  - D. Nucleosome depletion
4. Restriction endonucleases recognize palindromic sequences in DNA. Which of the following sequences would be recognized by a restriction enzyme? (Check all that apply.)
  - A. 3'-TAGCTA-5'
  - B. 5'-CGATTC-3'
  - C. 5'-GAATTC-3'
  - D. 5'-GAGCTC-3'
5. The leucine zipper motif functions to
  - A. anchor transcriptional activator proteins to the enhancer sequence
  - B. release leucines from misfolded transcription factors
  - C. integrate leucines into newly translated transcriptional activators
  - D. mediate the physical association of two polypeptides.
  - E. bring initiator tRNAs to small ribosomal subunit
6. Restriction endonuclease *NotI* recognizes eight nucleotide sequence (5'GCGGCCGG3'). Approximately how many restriction fragments would result from the complete digestion of the human genome (3 billion bases) with this restriction enzyme? Assume that the human genome has an equal proportion of all four nucleotides.
  - A. ~375 million fragments
  - B. ~12 million fragments

- C. ~70000 fragments
- D. ~46000 fragments
- E. 3 million fragments

7. Let's suppose you generated a mutation in the gene that encodes (makes) glucocorticoid receptor. In this mutant, the receptor protein is produced, and its hormone binding activity is not affected, meaning it binds to the hormone normally. However, in this mutant glucocorticoid inducible genes are not expressed in the presence of the hormone, glucocorticoid. Knowing the mechanism by which glucocorticoid induces gene expression, list **two possible** defects in the receptor that may explain why transcription cannot be activated.

8. What are microRNAs (miRNAs)? How are they produced? Describe two mechanisms by which they regulate gene expression.

9. What are the three features of a plasmid cloning vector that allow researchers to insert a piece of foreign DNA and make more copies of it in E. coli? Explain the purpose of each of these features.

10. What is a recombinant vector? How is a recombinant vector constructed? Explain how X-Gal is used to identify recombinant vectors that contain a segment of foreign DNA.

11. How are dideoxynucleotides (ddNTP) structurally different from deoxyribonucleotides (dNTPs)? How does this structural difference make ddNTPs valuable in Sanger's method of DNA sequencing?

12. The following DNA sequence (200 bp) is from a region of a eukaryotic organism. You want to clone (make billions of copies) this 200 bp DNA fragment in a test tube using polymerase chain reaction. You have the DNA from this organism, dNTPs, and DNA polymerase. All you need is to order forward and reverse primers from a DNA synthesis company.

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      20      40      60      80      100
5' ATGGCGGCGGCGGCGGACGCGCGGCGCTTCGCGGTGTCGCCGAGCTAGATATTGCGCAGATACTGAAGGAGGCTCAACAACGATGGTTGCGTCCTGCTGAAATTTGCGA 3'
3' TACCGCGCGCGCGCGCTGCGCGCGCGGAAGCGCCAGCACGGCGTCGATCTATAACGCGTCTATGACTTCCTCCGAGTTGTTGCTACCAACGCAGGACGACTTTAAACGCT 5'

      120      140      160      180      200      220
5' AATACCTTAAAACTACAAAAGTTTCGTATTGCACCAGAACCCAAACAGGCCTCAAAGTGGCTCGCTTTTCTCTATTTGATCGGAAAGTATTGAGATACCTTTAGGAAGG 3'
3' TTATGAATTTTGTATGTTTTCAAAGGCATACGTGGTCTTGGTGGTTTGTCCGAGTTTCACCGAGCGAAAAGGATAAAGTGCCTTTTCACTAACTCTATGAAATCCTTCC 5'

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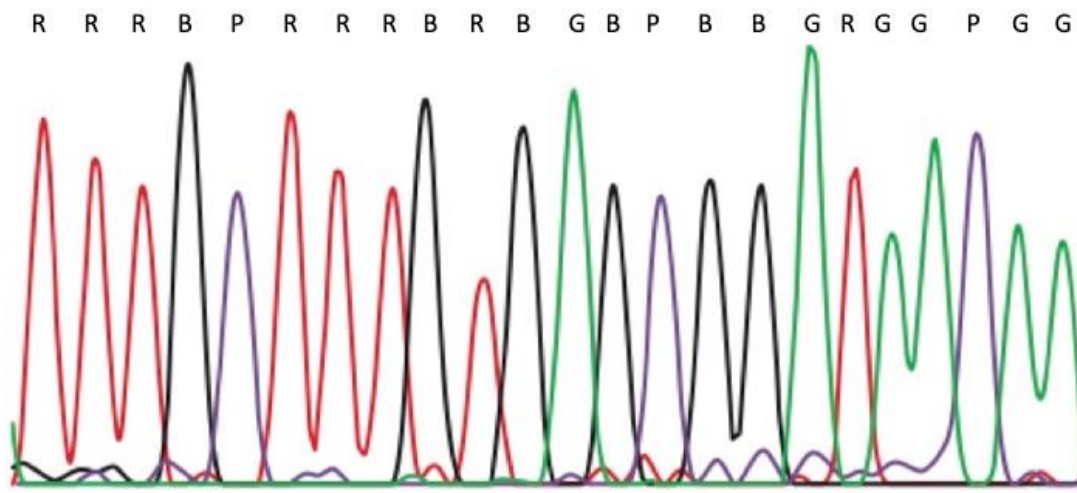
Which of the following sequences would you order for a forward and a reverse primer?

- A. 3' ATGGCGGCGGCGGCGGACGC 5'
- B. 3' TACCGCCGCCGCCGCCTGCG 5'
- C. 5' ATGGCGGCGGCGGCGGACGC 3'
- D. 5' TACCGCCGCCGCCGCCTGCG 3'
- E. 3' TATTGAGATACTTTAGGAAGG 5'
- F. 3' ATAACCTCTATGAAATCCTTCC 5'
- G. 5' TATTGAGATACTTTAGGAAGG 3'
- H. 5' ATAACCTCTATGAAATCCTTCC 3'

13. In Sanger sequencing, which of the following nucleotides are used?

- A. Color-coded deoxynucleotide triphosphates
- B. Color-coded deoxynucleotide triphosphates and normal (not tagged with color) dideoxynucleotide triphosphates
- C. Color-coded dideoxynucleotide triphosphates and normal (not tagged with color) deoxynucleotide triphosphates
- D. Color-coded normal deoxynucleotide triphosphates and 3'OH blocked deoxynucleotide triphosphates
- E. Color-coded 3'OH blocked deoxynucleotide triphosphates

14. You performed a Sanger sequencing reaction and obtained the following trace of DNA sequence. In this figure, A = green, C = purple, G = black, T = red. The height of the peaks is unimportant. The smallest DNA molecule is at the left of the trace.

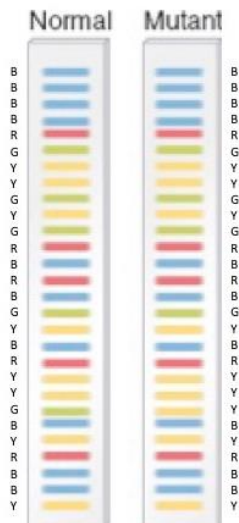


What is the sequence of the **DNA synthesized** in the sequencing reaction?

- A. 3' TTTGCTTTGTGAGCGGATAACAA 5'
- B. 5' TTTGCTTTGTGAGCGGATAACAA 3'
- C. 3' AAACGAAACACTCGCCTATTGTT 5'
- D. 5' AAACGAAACACTCGCCTATTGTT 3'

15. DNA sequencing can help to identify mutations in genes. The following data are derived from an experiment in which a normal gene and a mutant gene have been sequenced.

G = Yellow  
A = Green  
T = Red  
C = Blue



- i) What is the position of the mutated nucleotide?
- ii) What is the substituted nucleotide in the mutant?